



University
of Glasgow

Power Electronics

DC-AC Converters (Inverters)

逆变器

Dr Atif Jafri

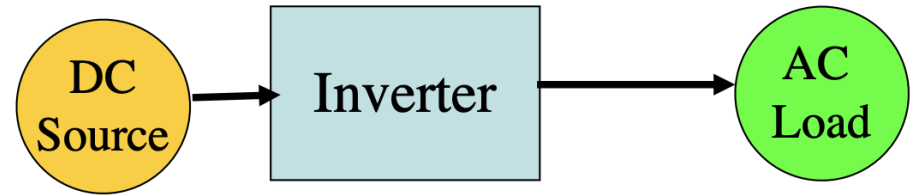
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Please read **Chapter 8** in the textbook

Scheme of Study

- Fundamental
- Types of Inverters
- Half-bridge Inverter
 - Ideal Circuit Diagram
 - Operation
 - Output Waveform
 - Harmonics
 - Practical Circuit Diagram
 - Avoiding shoot through

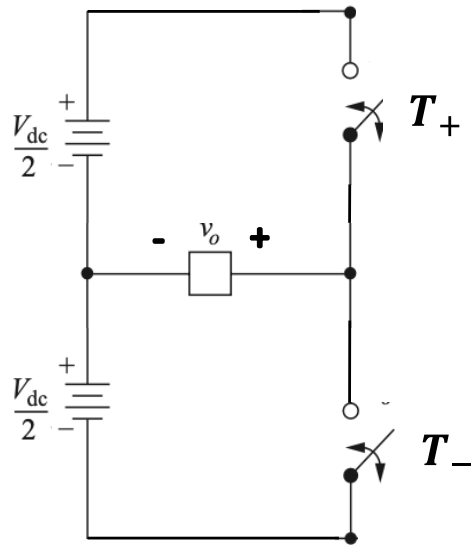
Fundamentals



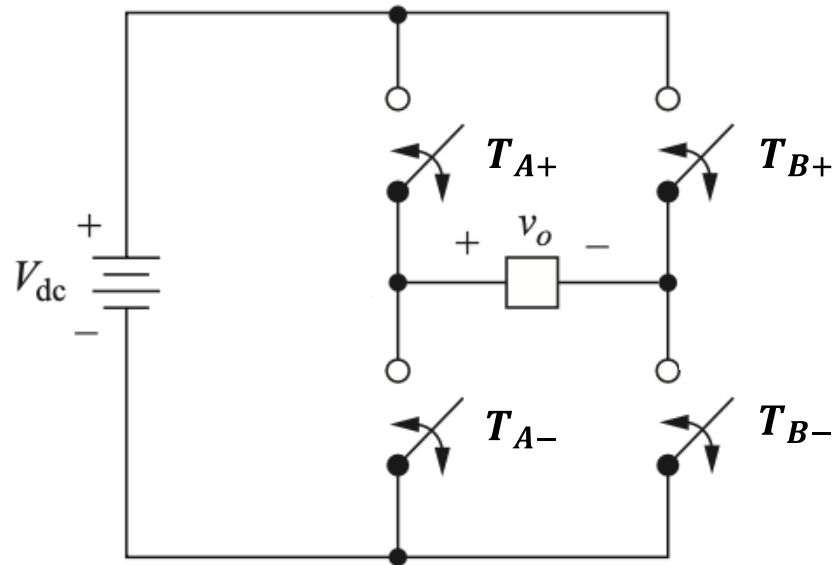
- DC-AC Converters are known as inverters
- Role is to convert a DC signal to AC
- Ideally, output should be sinusoidal.
- In reality, these are non-sinusoidal and contain harmonics
- This is fine for low and medium power applications
- Divided into two main types
 - Single Phase
 - Three Phase
- Semiconductor devices typically used
- The basic building block is the 'Bridge' circuit

Inverter Types

- Half and Full Bridge



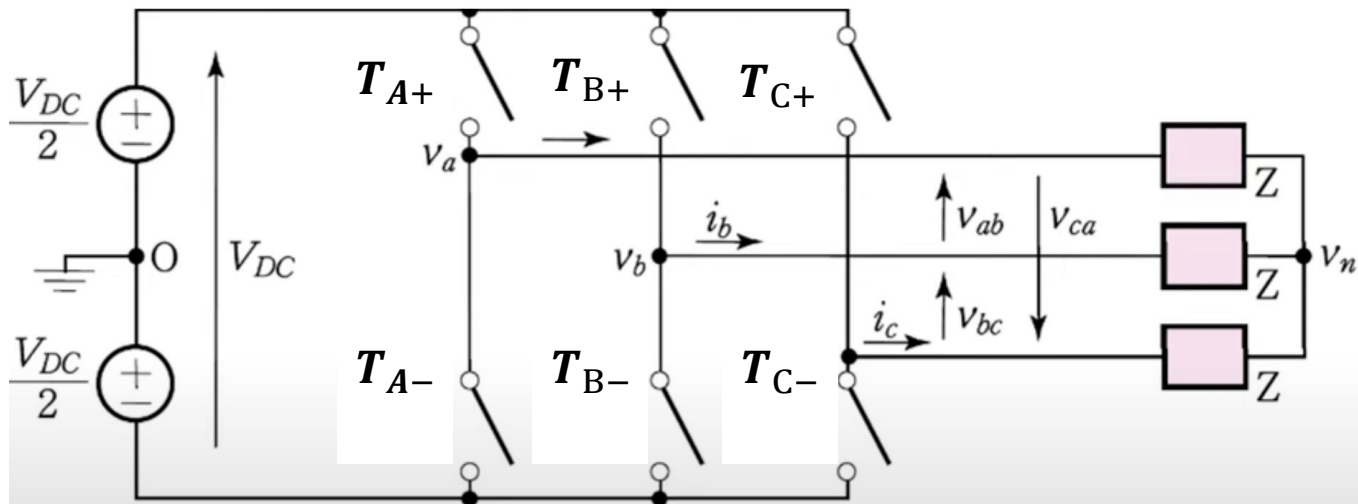
Half Bridge Inverter (Ideal)



Full Bridge Inverter (Ideal)

Inverter Types

3-Leg (For 3-phase inverter)



3-Leg Inverter (Ideal)

Schemes of Inverters to study

Phase Output	Bridge	Switching Schemes	• Ideal Circuit Diagram • Operation • Output Waveform • Harmonics • Practical Circuit Diagram • Avoiding shoot through
Single Phase	Half	Square Wave	
		Pules Width Modulation	
	Full	Square Wave	
		Pules Width Modulation (Bipolar & Unipolar)	
Three Phase	3-Leg	Square Wave	
		Pules Width Modulation	

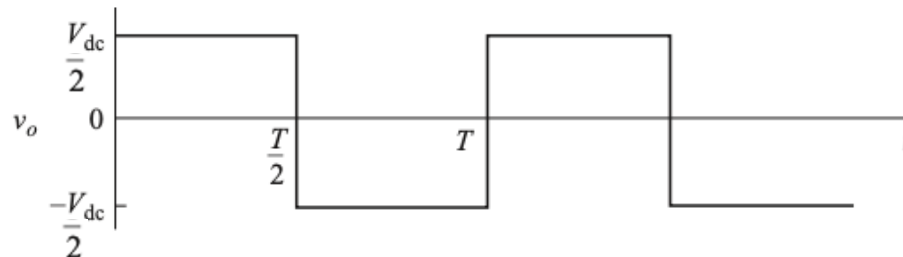
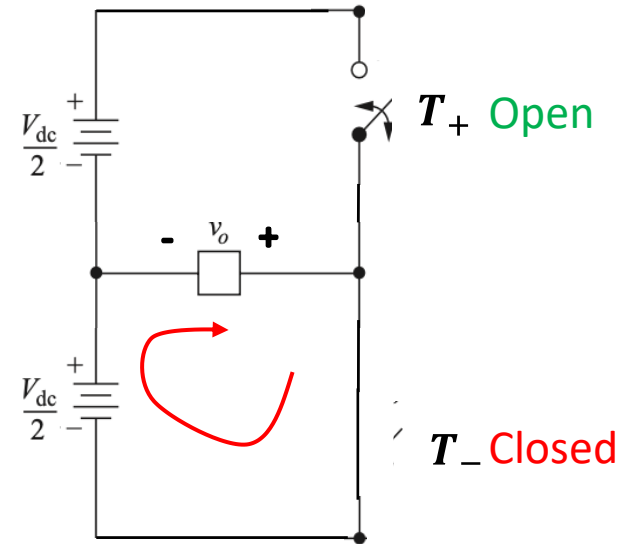
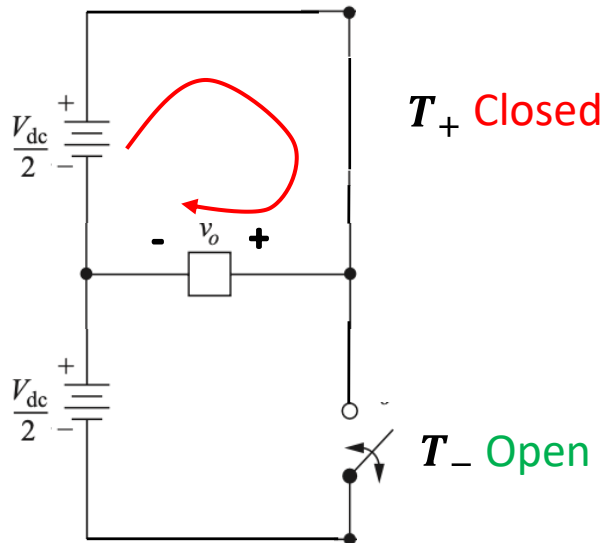
Schemes of Inverters to study

Phase Output	Bridge	Switching Schemes	• Operation in different switching schemes • Output Waveform • Harmonics • Practical Circuit • Avoiding shoot through
Single Phase	Half	Square Wave	
		Pules Width Modulation	
	Full	Square Wave	
		Pules Width Modulation (Bipolar & Unipolar)	
Three Phase	3-Leg	Square Wave	
		Pules Width Modulation	

Operation

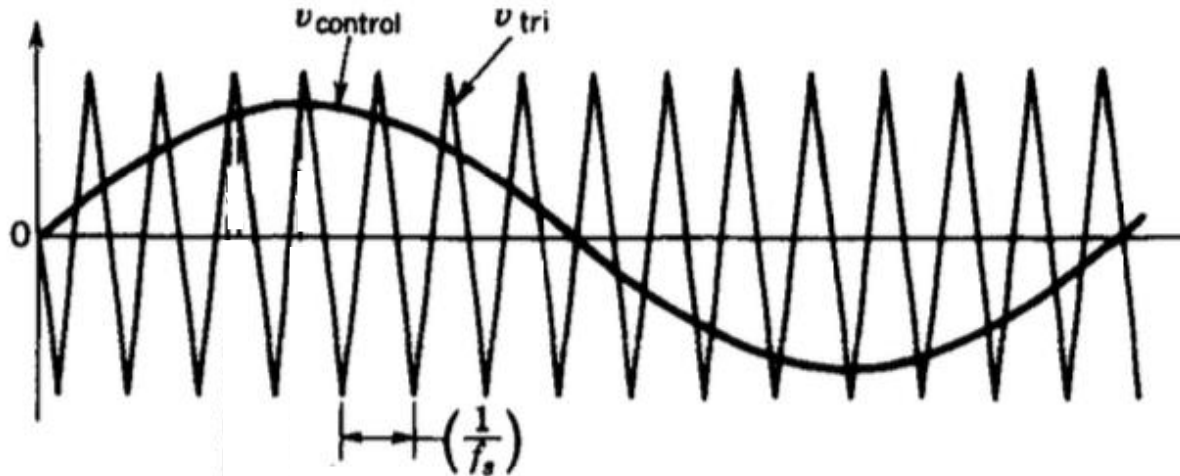
- Square Wave Switching

Half Bridge Inverter (Ideal)



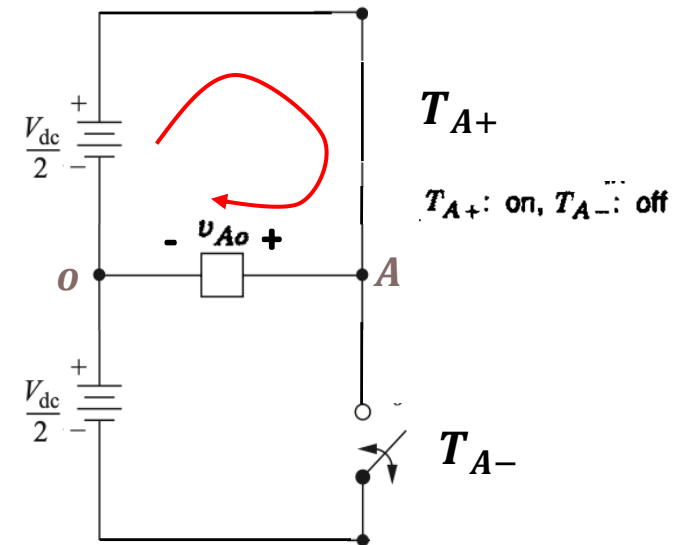
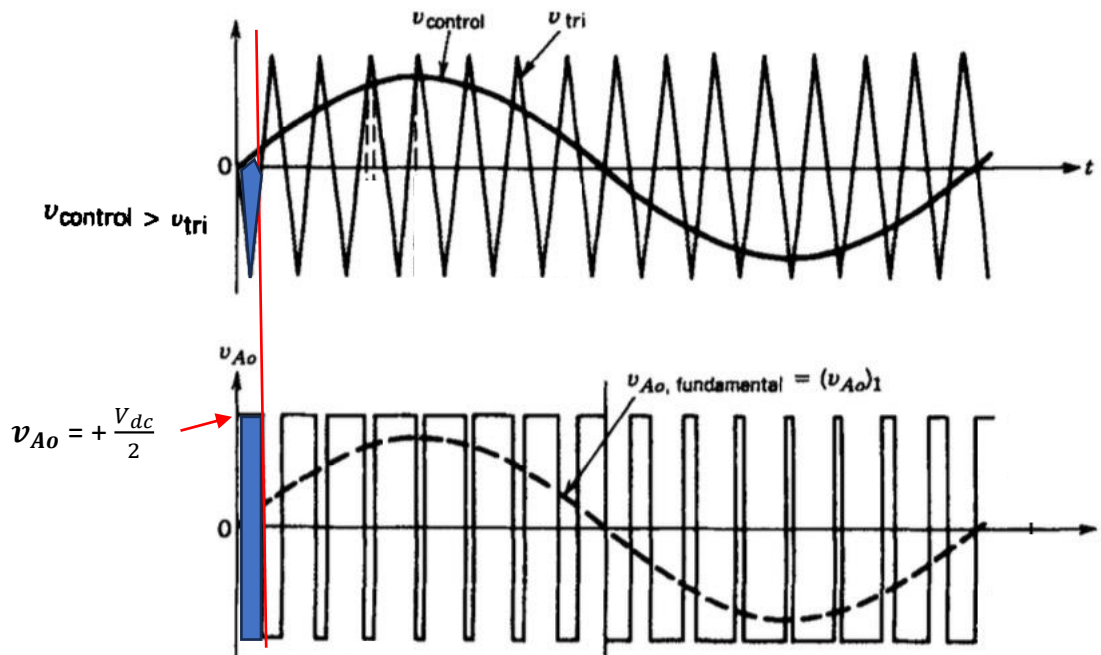
Operation

- Pulse Width Modulation (PWM) Switching
 - Is a type of switching technique.
 - Use two signals called
 - v_{tri} (Triangular Signal)
 - Has frequency f_s (called switching frequency)
 - $v_{control}$ (Control Signal)
 - Has frequency f_1 (desired fundamental frequency)



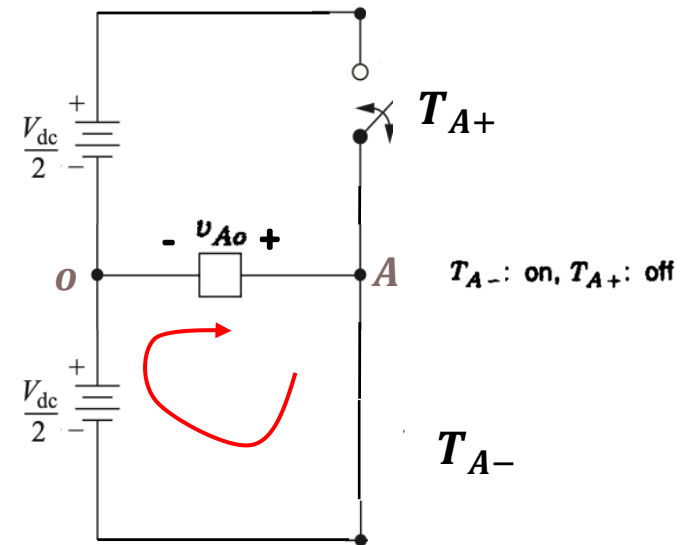
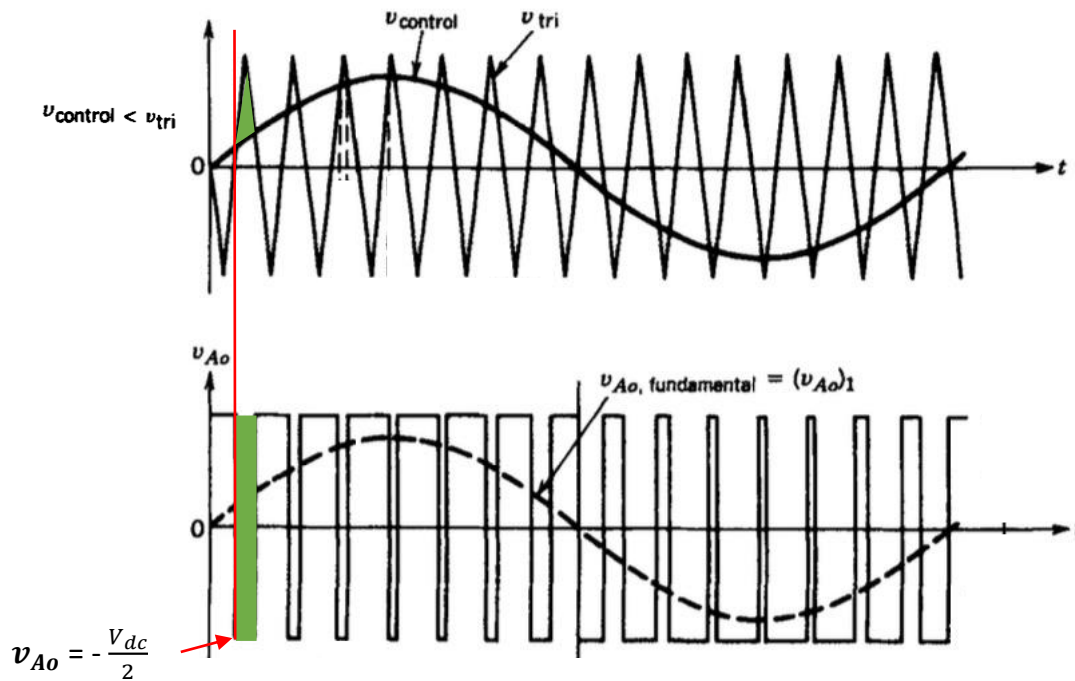
Operation

- Pulse Width Modulation (PWM) Switching
 - v_{tri} (Triangular Signal) is compared with $v_{control}$ (Control Signal)
 - Based on comparison result different switches are made closed (ON) and open (OFF)
 - Output voltage swings between positive and negative voltages



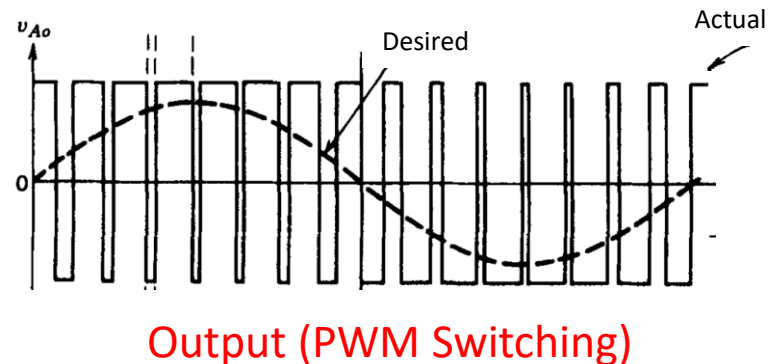
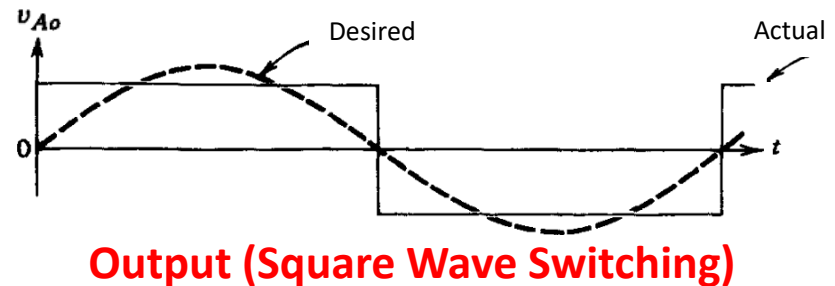
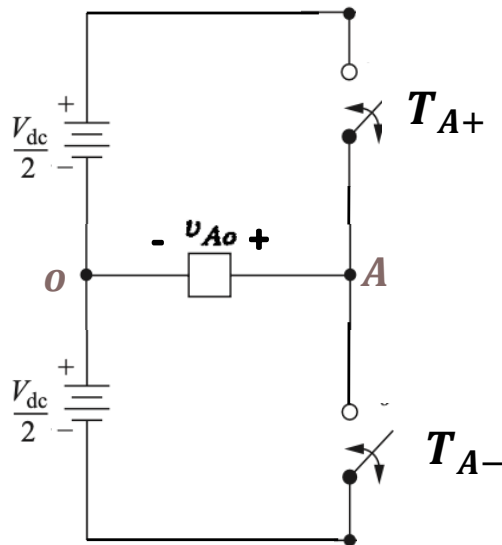
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Harmonics

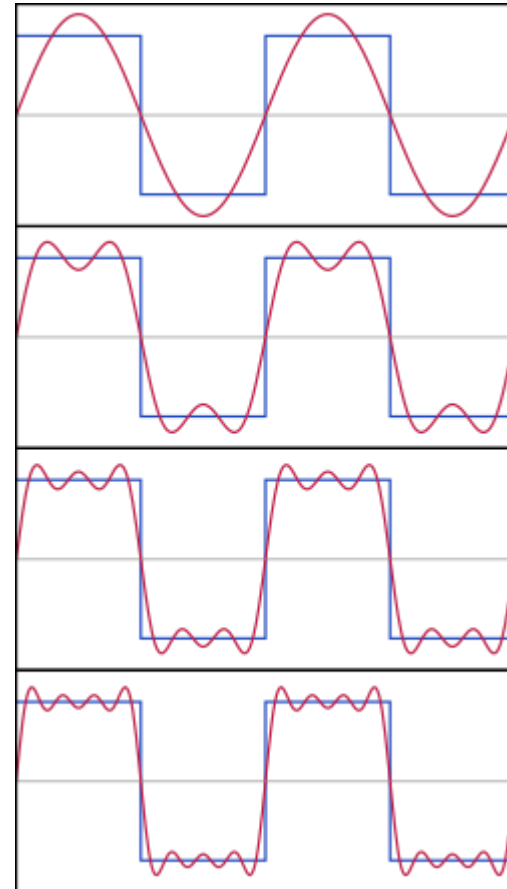
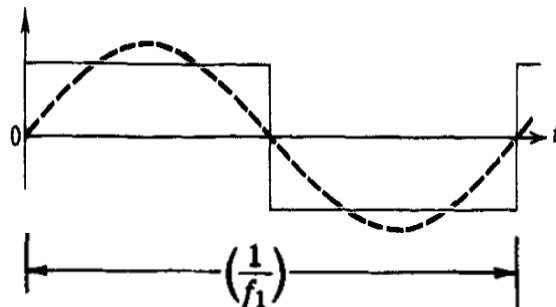
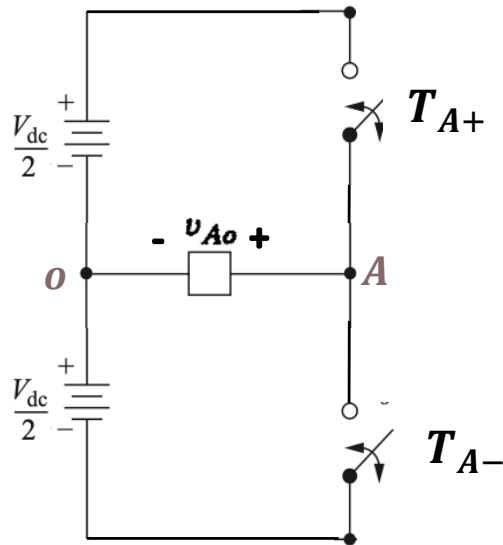
- What are harmonics?
 - Pure Sinusoid is required.
 - The output is different both in square-wave and PWM switching
 - There is need to analyze the **presence of other sinusoids** having **frequencies multiple of desired sinusoid**, **called harmonics**, available in output.



Harmonics

- Harmonics in Square Wave Switching

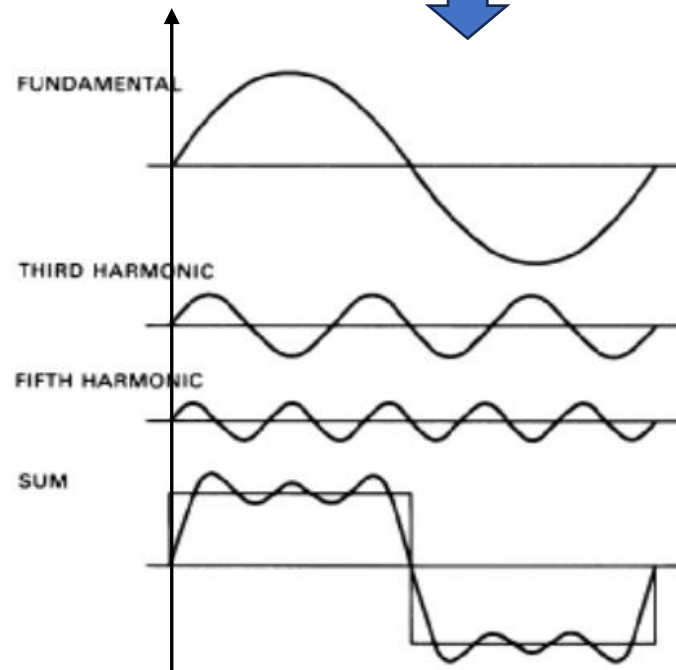
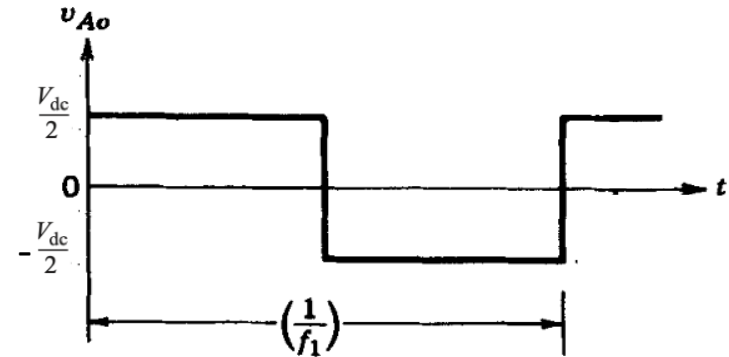
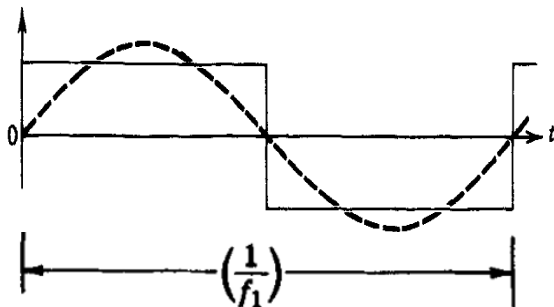
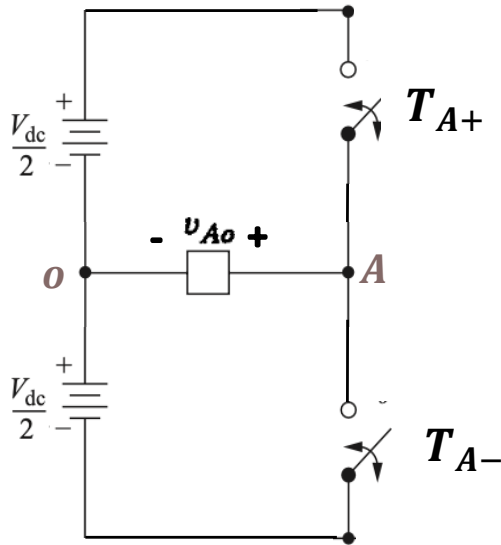
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Harmonics

- Harmonics in Square Wave Switching

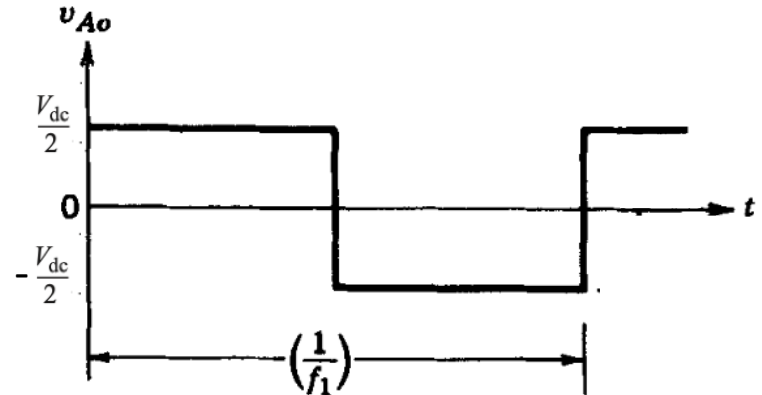
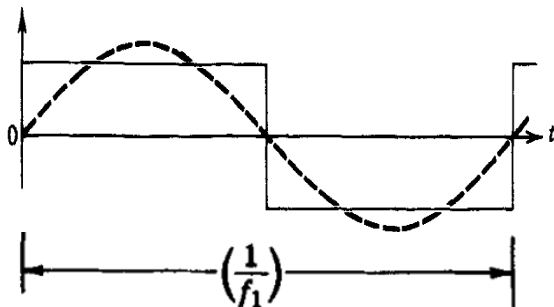
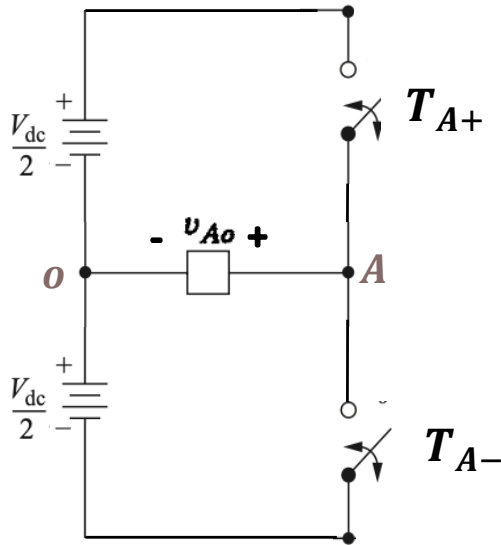
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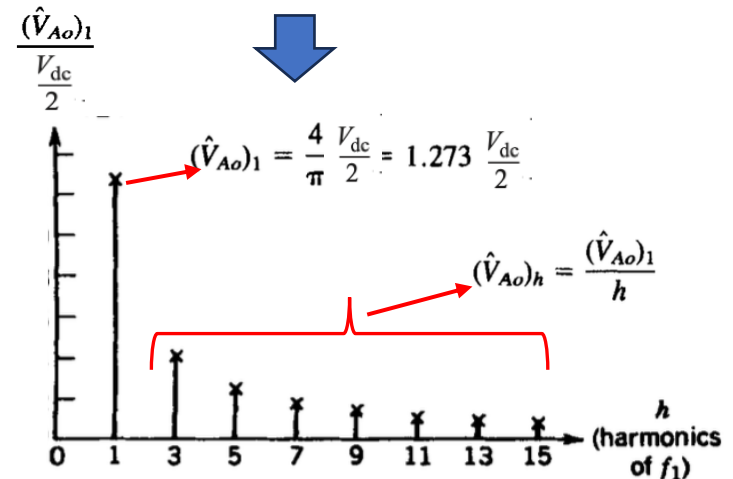
Harmonics

- Harmonics in Square Wave Switching

- There is need to analyze the **presence of other sinusoids** having **frequencies multiple of desired sinusoid, called harmonics**, available in output.



Fourier Series



Harmonics

- Harmonics in PWM Switching

- To analyze “*harmonics*” for PWM switching we need to understand two new terms:
 - **Amplitude Modulation Ratio** and **Frequency Modulation Ratio**.

$$m_a = \frac{\hat{V}_{\text{control}}}{\hat{V}_{\text{tri}}}$$



Amplitude Modulation Ratio

$$\hat{V}_{\text{control}}$$

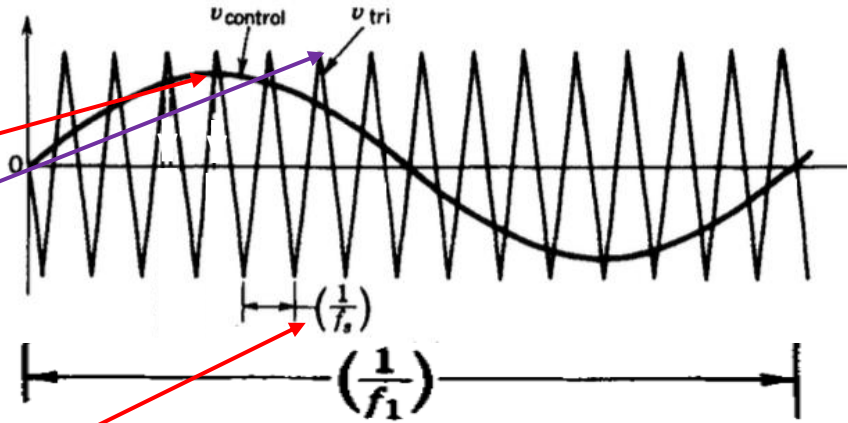


Peak Value of Control Signal

$$\hat{V}_{\text{tri}}$$



Peak Value of Triangular Signal



$$m_f = \frac{f_s}{f_1}$$



Frequency Modulation Ratio

$$f_s$$



Frequency of triangular signal
also called switching frequency

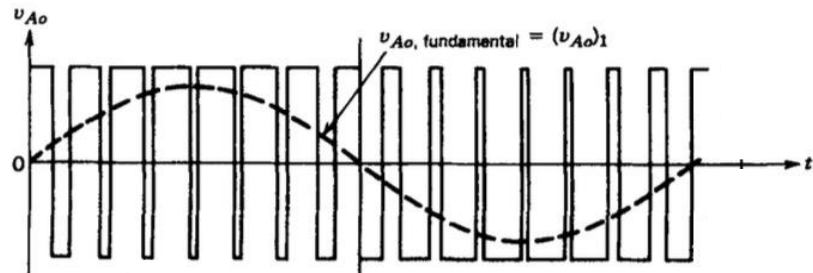
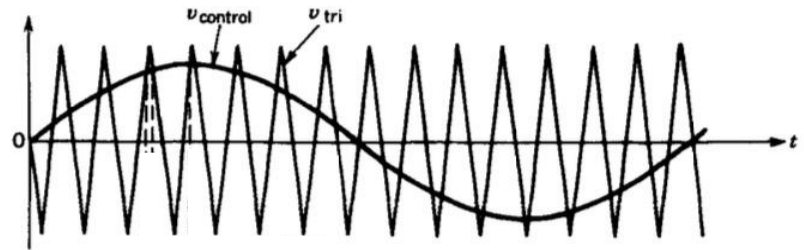
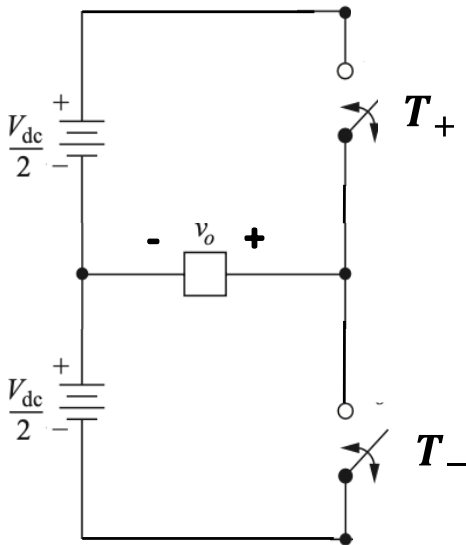
$$f_1$$



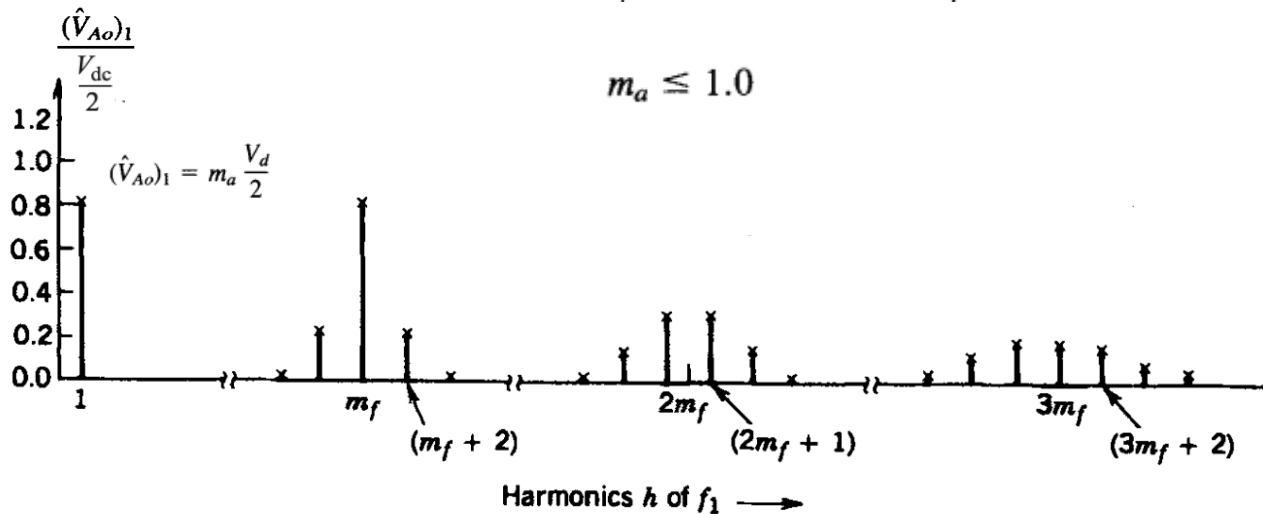
Frequency of control signal also
called desired fundamental frequency

Harmonics

- Harmonics in PWM Switching

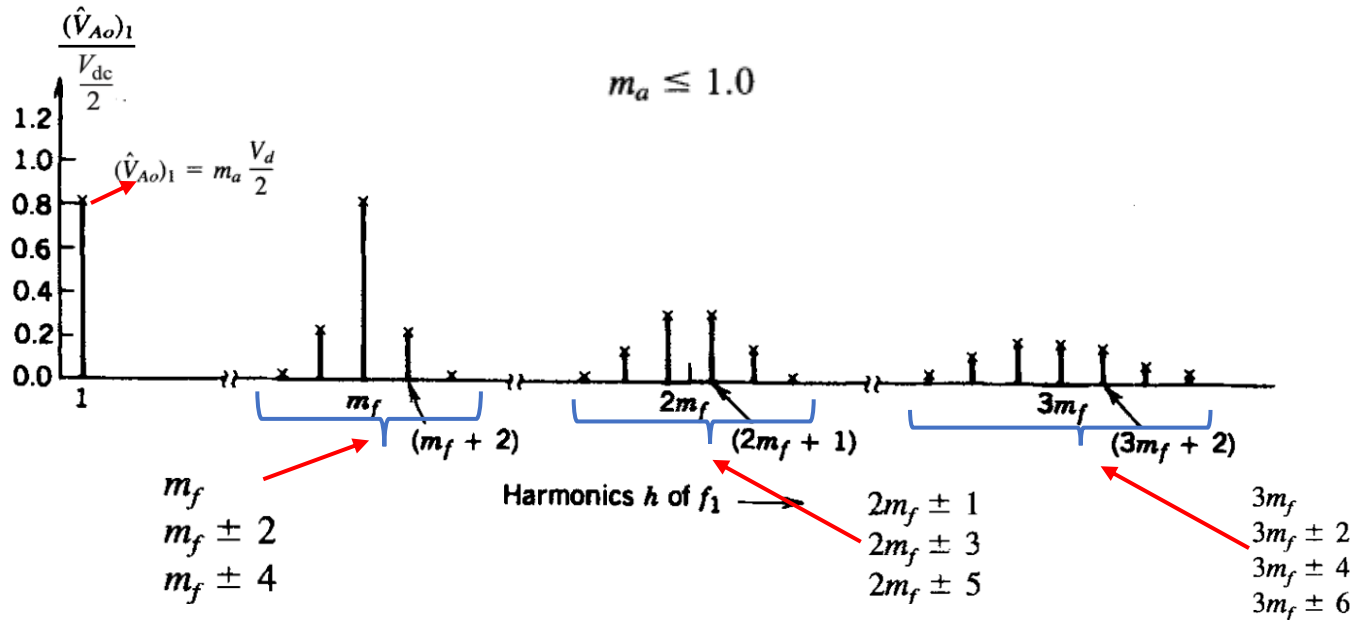


$$m_a \leq 1.0$$



Harmonics

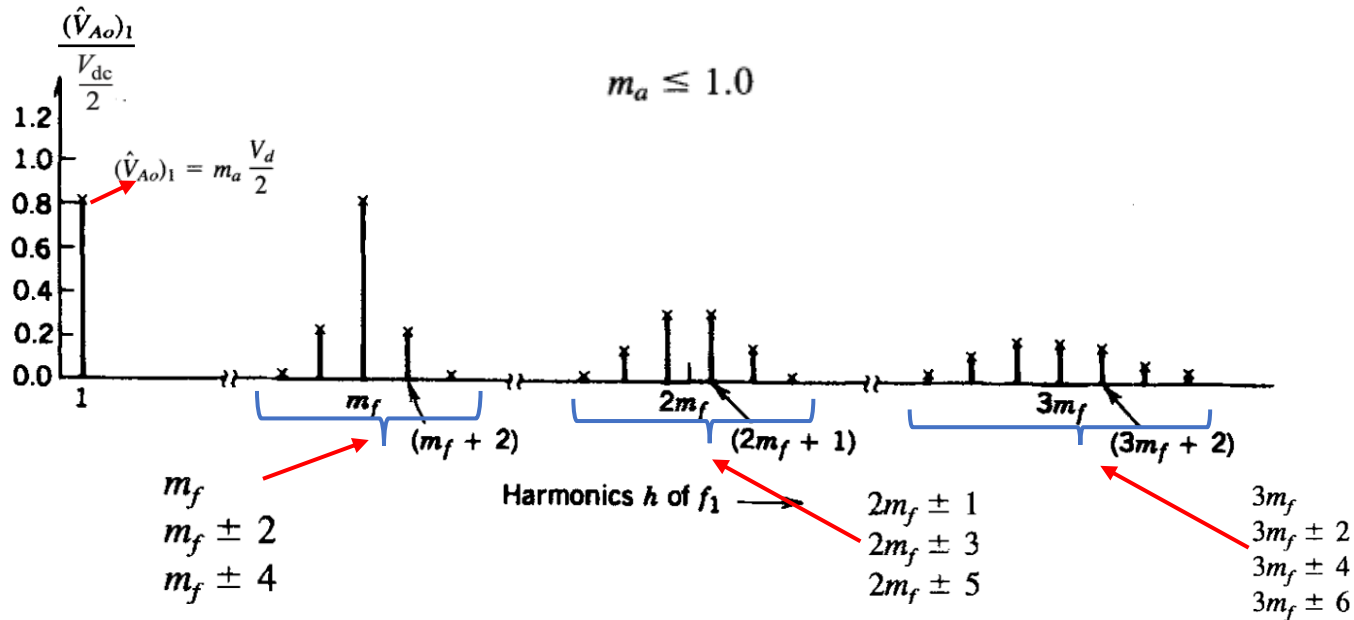
- Harmonics in PWM Switching



1. The peak amplitude of the fundamental-frequency component $(\hat{V}_{Ao})_1$ is m_a times $\frac{V_d}{2}$.
2. The amplitude of the fundamental-frequency component of the output voltage varies linearly with m_a (provided $m_a \leq 1$).
3. The harmonics in the inverter output voltage waveform appear as sidebands, centred around the switching frequency and its multiples, that is, around harmonics m_f , $2 m_f$, $3 m_f$, and so on.

Harmonics

- Harmonics in PWM Switching



4. Theoretically, the frequencies at which voltage harmonics occur can be indicated as:

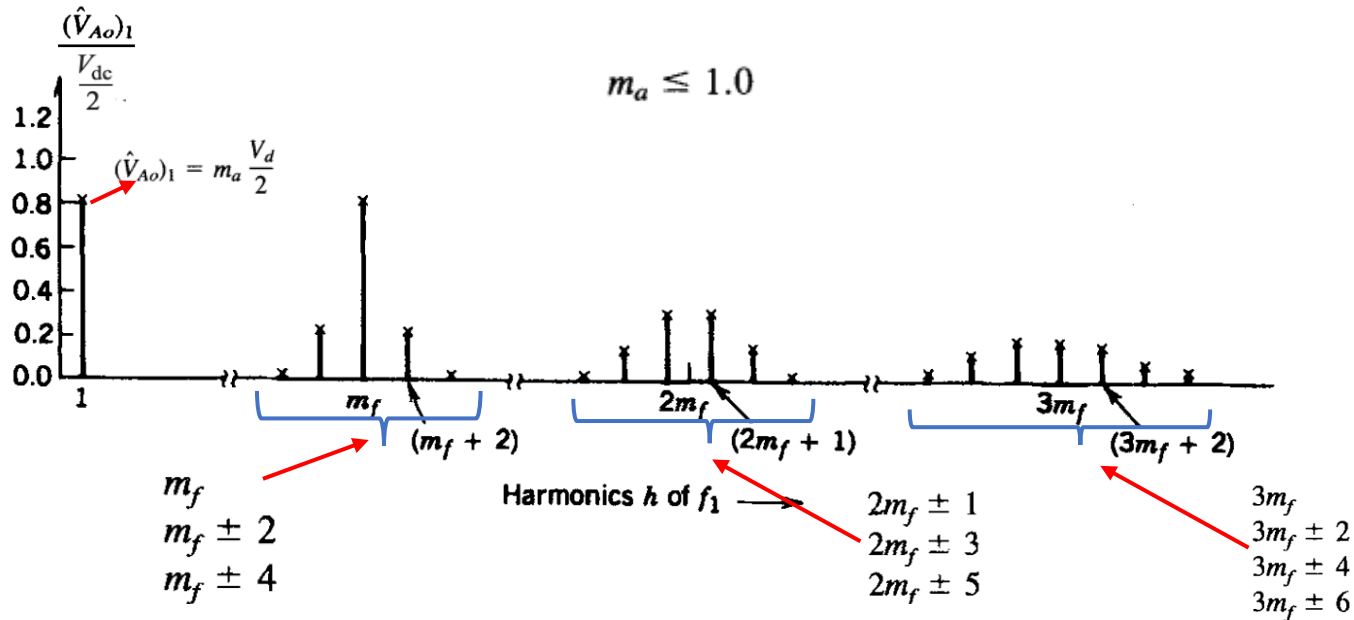
$$f_h = (jm_f \pm k)f_1$$

5. the harmonic order h corresponds to the k^{th} sideband of j times the frequency modulation ratio m_f

$$h = j(m_f) \pm k$$

Harmonics

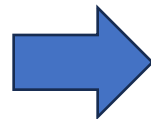
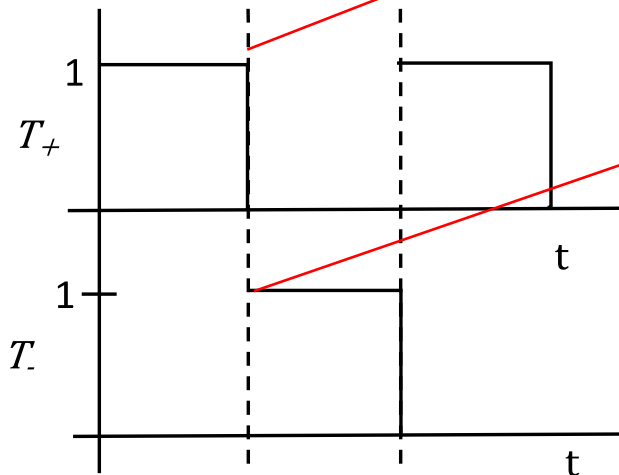
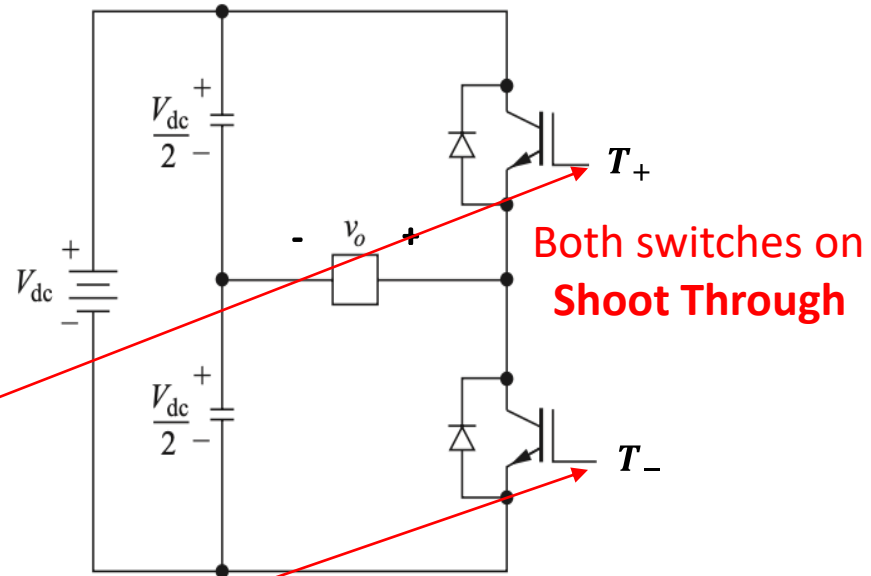
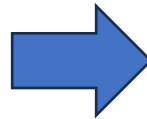
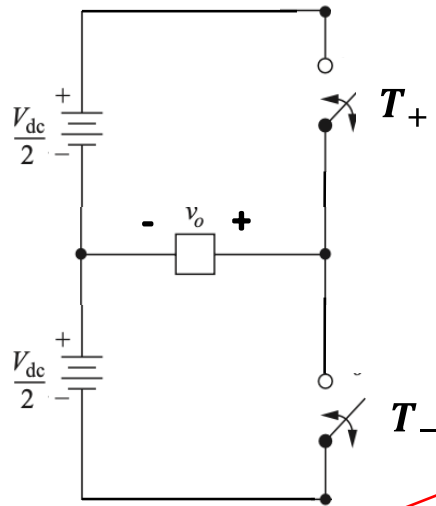
- Harmonics in PWM Switching



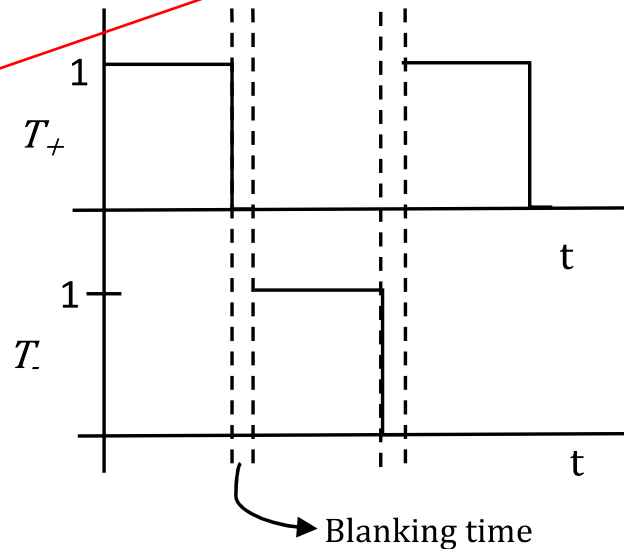
6. For odd values of j , the harmonics exist only for even values of k . For even values of j , the harmonics exist only for odd values of k . $f_h = (jm_f \pm k)f_1$

Practical Half Bridge Inverter

- Made through real components
 - Square Wave Switching

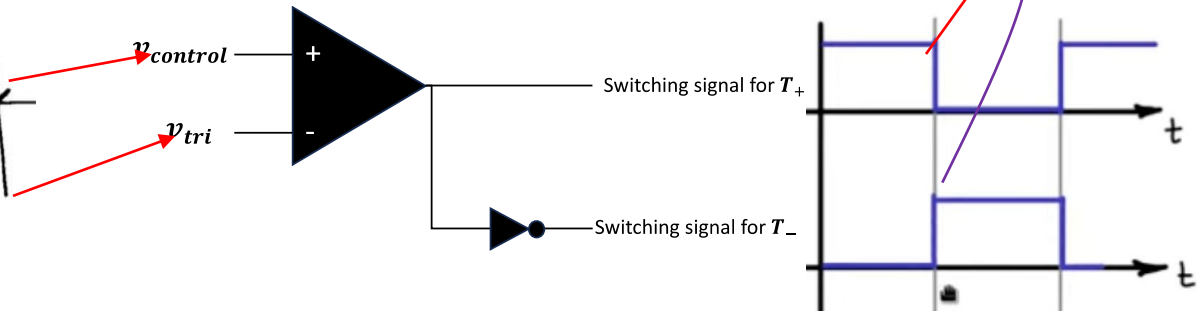
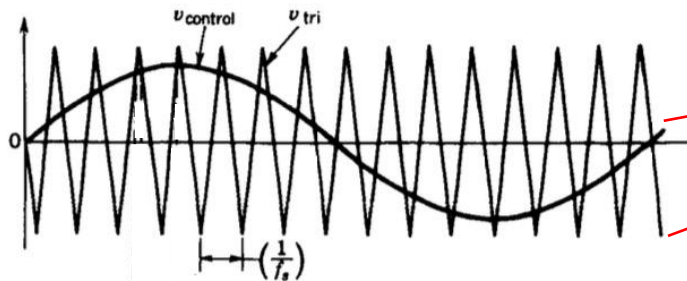
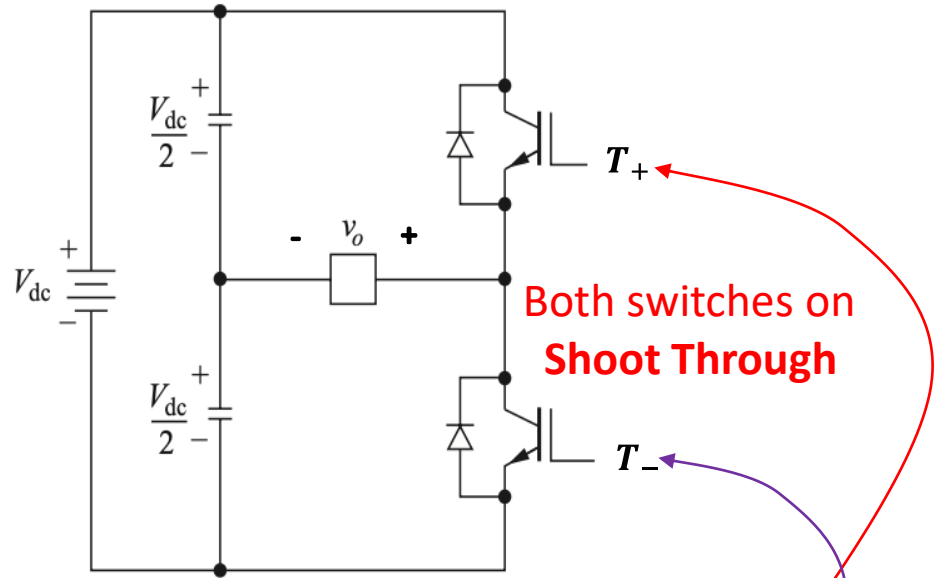
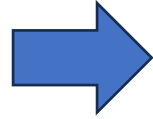
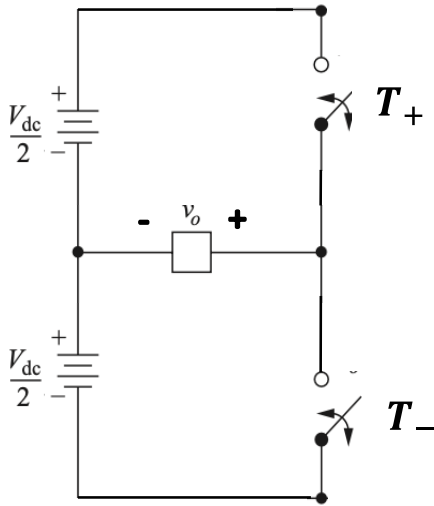


Avoiding
Shoot through



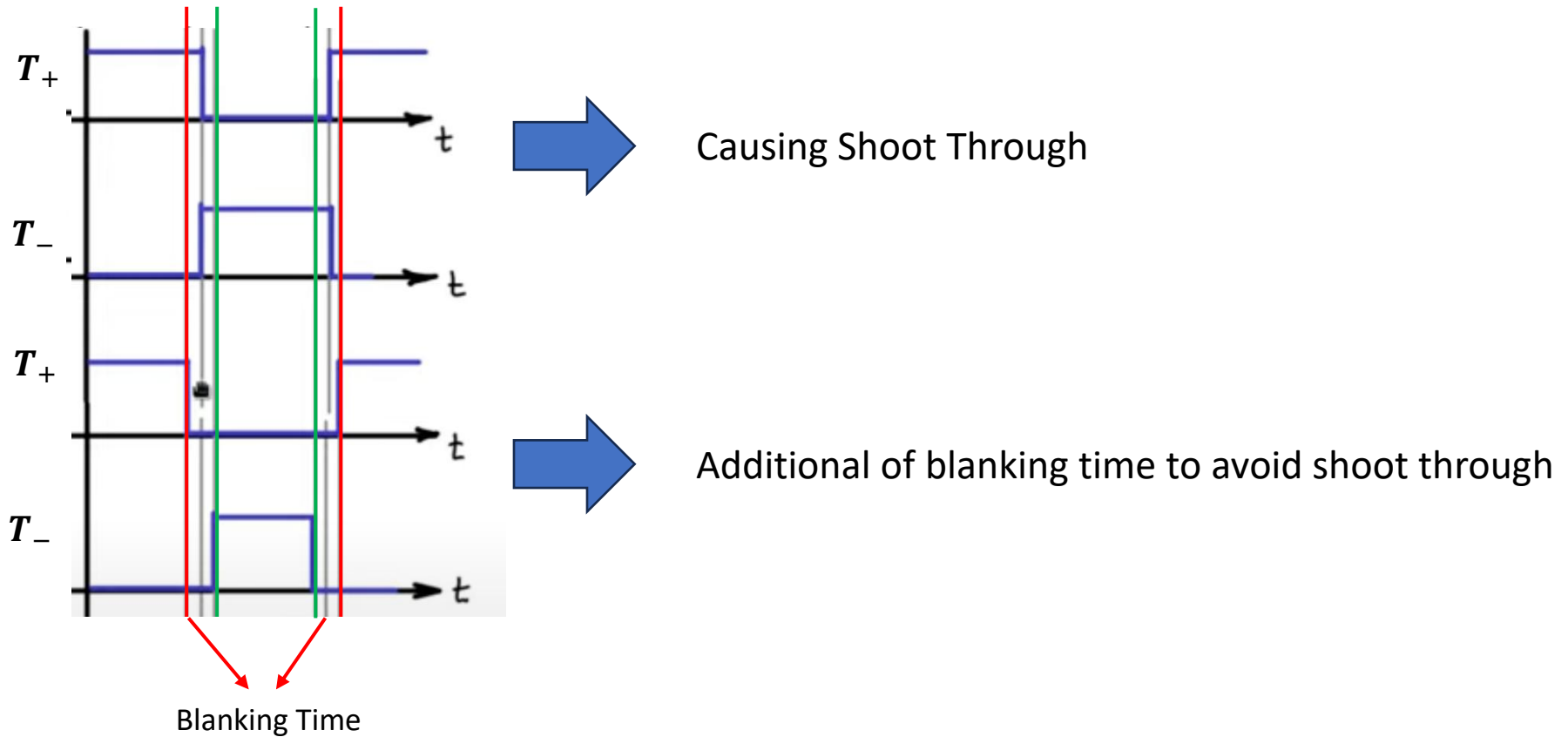
Practical Half Bridge Inverter

- Made through real components
 - PWM Switching



Practical Half Bridge Inverter

- Made through real components
 - Avoiding shoot through in PWM Switching



Avoiding Shoot through

